SOL v3.8 — Requirements-to-Evidence Matrix

Master specification ‘v3.7 Completion, v3.8 Major Upgrade & v4.0 Readiness’, section by section

2026-10-12

Status vocabulary.

- **Done-verified:** code, an automated test and image or metric evidence.
- **Done-software:** code plus software-WebGPU evidence, with no hardware check.
- **Partial**, **Open**, **Blocked**, **Not started**, **Doc**.

Evidence references. Image names refer to the QA evidence package (`QA_IMAGES/<set>/<name>.png` plus a JSON sidecar). Report names are abbreviated: V37, V38, AUD (scientific and asset audit), VQA (visual QA), PERF and HAND (master hand-off).

§	Requirement	Status	Evidence / where	Remaining
2	Desktop Chrome, native WebGPU only, no WebGL	Done-verified	`core/gpu.ts` probe; no-WebGPU error screen `final-ui5/load/load-error-no-webgpu`	—
3	No arbitrary quality ceiling	Done-software	no caps added; optimisations are worker/caching only (PERF §2)	hardware tiers
4	Honesty, evidence classes	Done	truth labels in asset manifest, inspector “Size source”, “Model” badge, enhancement labels	—
6–7	Baseline recovered, audited, reproduced	Done-verified	tag `v3.7.0` = upload (2,158 files identical); tsc/tests/build reproduced (V37 §2)	—
8	Preserve verified foundations	Done-verified	regression audit + v3.7.1 hunk check (RELEASE_VERIFICATION)	—
9	Outstanding-work register, classified	Done	V37 §3	—
10	Venus terminator	Done-software	`final/venus-terminator` vs `v370-A` (−9.25 → −1.32 EV); cloud-top flight	real-GPU look
10	Jupiter backlit rings	Done-software	`final/jup-backlit(-sci)`	—
10	Io plumes	Done-software (REPRESENTATIVE look)	`final/io-pele-limb-*`, `io-tvashtar-limb`	time-variable activity
10	Enceladus plumes	Done-software	`final/enceladus-plume-*` (0 clipped px)	—

§	Requirement	Status	Evidence / where	Remaining
10	Other regressions	Done	DEF-38-001…013 (V38 §9)	—
11	Ring completion (Uranus eccentric, tiers, Pan/Encke)	Partial	`final/ura-*`, `seq-ura-epsilon-precession_*`, `moon-pan`; test innermoons38, rings371	R5 volumetric, wakes, spokes, propellers (Open)
12	Atmosphere completion	Partial	V37 §5; DEF-38-001 fix	volumetric clouds (CLD-001 Open)
13	v3.6 obligations	Partial	LUT worker done; terrain patch worker not done	terrain worker
14	Moon inventory + appearance policy	Partial	`docs/inventory/moons-v371.json`, `satSizes.json`, `final/moon-*` (24 views)	controlled small-moon mosaics
15	v3.8 release title / workstreams	Done	V38 §1	—
16–17	Far-Sun regimes (photosphere → disc → subpixel → star)	Done-software	`final/sunfar-*` (13 distances × modes), `seq-sunfar_f00–15`, Sun at 1 ly / 1 pc / Proxima distance	real-GPU bloom
18	Far-Sun QA set	Done-software	VQA §2	4K captures (1600×900 used)
19	Lighting system	Partial	metering fixes; planetshine unchanged	BRDF upgrades, GI
20–21	Galactic architecture, source data	Done (foundation)	Gaia DR3 20-pc refinement; GCNS + Hipparcos; G→V photometry (DEF-38-013)	Gaia bulk > 100 pc
22	Catalogue streaming	Done-software	journey metadata: shards 0 → 39 → 85 resident; SHA-256-verified	bulk tiles
23	Reference frames	Done-verified	`contracts/frames.ts`, tests (Sgr A*, NGP)	—
24	First interstellar journey (Earth → Proxima → back)	Done-software	`final-journey/*` (21 frames, finite, monotonic), Sun from Proxima, α Cen from Proxima	hardware smoothness
25	Milky Way background + structural model	Partial (model + card; panorama unchanged)	`milkyWayModel.ts` + tests; Galactic card in UI shots; `final-gal/*`	rendered 3-D Galaxy (v3.9)
26	Deep-sky reference objects	Done (markers)	9 objects, `deepSky.ts`	volumetric rendering
27–29	Visual identity, tiers O1–O4, no repetition	Partial	size-dependent geology, families, AUD §4	mosaics
30	Textures and materials	Partial	asset manifest (1,513 assets)	KTX2, 3 NC maps
31	Small bodies and shape models	Done-software (2 new)	Itokawa, Phoebe GLB (PDS)	more SBN models
32	High-resolution planetary surfaces	Not started in v3.8	v3.6 terrain unchanged	LROC/CTX imagery

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33	Improve, do not rebuild, the UI	Done	UI polished in place	—
34	Loading & intro	Done-software	`final-ui5/load/*` (7 stills + error)	—
35	Navigation & information	Done-software	search fix, star inspector, Galactic card, journeys	panel docking (UX-002)
36	Motion & transitions	Done-software	radial interstellar legs; reduced-motion honoured (ui38)	hardware motion review
37	Accessibility	Done-software	ui38 on release build: 230 controls, 0 unnamed, 0 < 24 px, 0 low-contrast; Tab/Esc/R/search checks	screen-reader pass with real AT
38–39	Travel stutter: reproduce + profile	Done-software	PERF §2–3, 3+3 runs	hardware traces
40–41	Optimisation without visual compromise	Done-software	workers, pre-smoothing, predictive prep; identical images (DEF-38-001 caught a precision loss, fixed)	KTX2
42	Real hardware validation	**Blocked**	no GPU in this environment	owner RTX 3050 run (guide in docs/qa)
43–46	v3.9 / v4.0 readiness	Doc	HAND §7–9	—
47–48	Sources and media rights	Done	AUD §1–2, CREDITS.md	3 NC maps
49–57	Still-image QA with metadata	Done-software	94 final views + 6 Galactic + journey 21 + UI 11 + loader 8, JSON sidecars	4K; terrain not re-captured
58–61	Six reports, MD + image-rich PDF	Done	`docs/reports/v3.8/`	—
62	Engineering verification	Done-verified	tsc 0 errors, 310/310 tests, assets 0 unmatched, build, clean-extract boot (RELEASE_VERIFICATION)	—
63–68	Acceptance gates	Scientific/functional/doc: met in software scope; visual: met with listed residuals; performance & UX: **not met until hardware validation**	VQA §10, PERF §6	hardware
69	Recovery & checkpointing	Done	git history, tags `v3.7.0`, branch `v3.7.1`, tag `v3.8.0`, WORKLOG	—
71–72	Releases	Done (owner-amended: single consolidated ZIP)	`sol-solar-system-v3.8-final.zip`	—
73	Final scope audit	Done	this matrix	—